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Are galaxies in compact groups special?

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ABSTRACT

Context. Compact groups combine the highest projected galaxy densities with low velocity dispersions, yet most of them lie inside larger groups. Whether their galaxies are special therefore depends on how they compare with the members of ordinary groups, not only with the field.
Aims. We test whether the galaxies of compact groups of four members (CG_4) differ from those of regular groups of four members (RG_4), from the four brightest members of regular groups ($Control_{4B}$), and from the projected cores formed by the brightest group galaxy and its three nearest projected companions ($Control_{4C}$).
Methods. For 62 non-split CG_4 systems we combine Galaxy Zoo 1 morphologies, stellar masses, specific star-formation rates, emission-line diagnostics, spectral indices, and seeing-corrected half-light radii. Each control is compared separately with covariate-adjusted models, galaxy- and group-level matching, and comparisons within shared host groups.
Results. The clearest difference is morphological. At fixed stellar mass, redshift, rank, and group properties, CG_4 galaxies are more often in the Galaxy Zoo 1 E class, especially their satellites, whose odds ratios are 2.63, 1.93, and 3.03 against $Control_{4B}$, $Control_{4C}$, and RG_4. Matched quartets show the same ordering: the excess of early-type-like satellites is significant against $Control_{4B}$ and RG_4, and has the same sign but is not significant against $Control_{4C}$ ($\Delta = 0.11$, $p = 0.083$). Within shared hosts, the CG_4 association is substantially attenuated after accounting for host-centric radius, although its interval permits a non-negligible residual association. Morphology accounts for part of the satellite-quenching contrast; simultaneous adjustment leaves a residual association against $Control_{4B}$ but not clearly against $Control_{4C}$ or RG_4. Other star-formation, colour, AGN-like, magnitude-gap, and size diagnostics show no comparably robust offsets.
Conclusions. In this population, compact groups of four galaxies host a population shifted towards early-type-like satellite morphologies. The quenching evidence that remains at fixed morphology is control-dependent and does not establish a compact-group-specific causal mechanism.

Key words. galaxies: groups: general – galaxies: evolution – galaxies: interactions – galaxies: star formation – galaxies: elliptical and lenticular, cD – methods: statistical

1. Introduction

Given their high apparent densities and low velocity dispersions (Hickson 1982; Hickson et al. 1992), compact groups appear to be ideal sites for the frequent tidal encounters and mergers that alter galaxy masses, luminosities, sizes, morphologies, and star-formation rates (e.g., Mamon 1992). Mendes de Oliveira & Hickson (1994) provided the classic report of an excess of early-type galaxies in compact groups relative to the field (building on the photometric morphologies of Hickson et al. 1989 and the merger-remnant expectations of Zepf & Whitmore 1993), in line with the broader morphology–density relation (Dressler 1980). Compact-group versus loose-group and field comparisons have since found broad differences in morphology and other galaxy properties, including the analysis of Coenda et al. (2012); multi-wavelength studies likewise report altered infrared, UV, star-formation, and stellar-population properties (Johnson et al. 2007; Tzanavaris et al. 2010; Bitsakis et al. 2011), and H I deficiency has been organised into an evolutionary sequence (Verdes-Montenegro et al. 2001).

The interpretation of such contrasts is nevertheless ambiguous. Projected selection can include chance alignments, and many Hickson-like systems need not be isolated bound entities: apparent compact groups may be transient alignments, dense substructures, or cores embedded in looser groups and richer haloes (Mamon 1986; Díaz-Giménez & Mamon 2010; Díaz-Giménez et al. 2020; Zheng & Shen 2021; Taverna et al. 2023). The relevant observational question is therefore not only whether compact-group galaxies differ from field galaxies, but whether they remain different from galaxies occupying regular-group analogues selected with comparable luminosity, redshift, and membership constraints. Recent structural studies find evidence for transformation in compact groups, especially non-isolated systems (Montaguth et al. 2023, 2025, 2026a,b); Montaguth et al. (2026b) compared non-isolated compact-group galaxies with surrounding host populations, but not with ordinary-group quartets reproducing compact-group membership, luminosity-ranking, or projected-core selection.

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In our previous work (Tricottet et al. 2025), we found that, once the direct consequences of the compact-group selection are accounted for, compact groups of four galaxies have group-scale properties broadly similar to those of regular groups of four galaxies and to the eligible cores of regular groups. Here we ask how the galaxy-population contrast changes across three ordinary-group comparison samples: genuine four-member groups (RG_4), the four brightest eligible members of richer groups ($Control_{4B}$), and BGG-centred projected cores ($Control_{4C}$), the closest analogue of a compact configuration. The incremental contribution is the joint use of these explicit comparator definitions, BGG–satellite separation, adjusted regressions, individual and quartet matching, and shared-host models with host-centric radius. This design tests how the interpretation changes with the comparator and with location inside the host; it does not propose a new evolutionary mechanism. Section 2 describes the samples, the classifications, and the statistical approach; Sect. 3 presents the results; Sect. 4 discusses their physical interpretation; and Sect. 5 concludes.

2. Data and classifications

2.1. Samples

We use the compact-group and control samples constructed in our previous study (Tricottet et al. 2025). Galaxy properties are taken from the SDSS DR12 catalogue of Tempel et al. (2017), while regular groups are drawn from the SDSS DR7 group catalogue of Lim et al. (2017); the two catalogues were matched through SDSS object identifiers to define the Lim–Tempel working catalogue. Compact groups are selected from the modified Hickson compact-group catalogue of Díaz-Giménez et al. (2018): reliable HMCGs with exactly four members, all present in the Lim–Tempel catalogue, satisfying the volume and BGG-luminosity limits $0.005 \leq z_{BGG} \leq 0.0452$ and $M_{r,BGG} \leq -21.81$. The original HMCG membership uses the Hickson three-magnitude concordance in apparent r magnitude, whereas the Lim-based control eligibility below uses the catalogue absolute magnitude; these conventions are not identical (retained CGs 173, 229, and 354 span 3.070, 3.063, and 3.182 mag in absolute r). In contrast to our previous study, we remove CGs classified as *split* following Zheng & Shen (2021), which reduces the compact-group sample from 78 to 62 groups. Following Zheng & Shen (2021), the remaining systems are *isolated* when the four members constitute their whole Lim–Tempel group, *predominant* when they contribute at least half of the host-group r-band luminosity, and *embedded* otherwise (6, 37, and 19 systems). [1]

The three control samples are built from Lim–Tempel groups satisfying the same redshift and BGG-luminosity limits, with at least four members and at least three satellites within 3 absolute magnitudes of the BGG, and serve different purposes (Fig. 1). RG_4 contains regular groups with exactly four members and is the most direct equal-membership comparison. $Control_{4B}$ contains the four brightest eligible members of each regular group and tests whether the CG_4 population resembles the luminous core of an ordinary group. $Control_{4C}$ contains the BGG and its three closest projected companions among members within 3 magnitudes of the BGG, reconstructed here from the eligible parent groups by applying the magnitude criterion before ranking companions by projected distance; it is the most compact-core-like analogue, but not a compactness-matched sample: the CG_4–$Control_{4C}$ overlap coefficient in $-\log\langle R_{ij}\rangle$ is 0.34, and 18% of CG_4 groups lie above the $Control_{4C}$ 95th percentile in compactness. A candidate control quartet is excluded when it contains a CG_4 galaxy, so no galaxy appears on both sides of a comparison; this does not exclude every other quartet drawn from a host that contains a different CG_4 member. Among the 56 non-isolated compact groups, 51 overlap their would-be host core and 46 are exact four-member identities (Appendix C). We also remove the systems associated with Lim group 3688, whose single outlying-redshift member ($z = 0.048$, against 0.032–0.034 for the other six) enters both the $Control_{4B}$ and $Control_{4C}$ quartets and inflates their velocity dispersions to $\simeq 2000\,\mathrm{km\,s^{-1}}$. Table 1 summarises the final samples; the $Control_{4B}$ and $Control_{4C}$ group counts differ because the luminosity-ranked and distance-ranked definitions retain different Lim groups after the compact-group exclusion.

Table 1. Final samples

Sample	Groups	Galaxies	$\langle R_{ij}\rangle$ (kpc)
CG_4	62	248	143 (114–177)
$Control_{4B}$	698	2792	447 (323–604)
$Control_{4C}$	703	2812	293 (187–436)
RG_4	56	224	430 (292–494)

Notes. $\langle R_{ij}\rangle$ is the median of the six pairwise projected physical separations in each quartet, derived from angular-diameter distances in the adopted Planck 2015 cosmology. Entries give the median and interquartile range over groups.

We use SDSS DR16 (Ahumada et al. 2020) to retrieve stellar masses, star formation rates (SFRs; Brinchmann et al. 2004; Kauffmann et al. 2003b), emission-line measurements, and Galaxy Zoo morphologies (Lintott et al. 2008, 2011); Appendix A lists the tables and joins. Throughout, 'SDSS non-AGN reference' denotes the AGN-filtered reference sample used for calibration and orientation: spectroscopic galaxies with $0.005 \leq z \leq 0.0452$, extinction-corrected Petrosian $r \leq 17.77$, a valid MPA-JHU stellar mass, and finite Galaxy Zoo vote fractions after the AGN filter. It is a field-dominated reference population, not a control sample. Projected physical separations are derived from angular-diameter distances in the adopted Planck 2015 cosmology (Planck Collaboration et al. 2016), as in Paper I; all distances quoted below are in kpc.

2.2. Classifications

Star-formation classes are based on SDSS total specific star-formation rates. The inherited DR12 group catalogues store $\log \mathrm{sSFR} = \texttt{sfr_tot_p50} - \texttt{lgm_tot_p50}$. In the MPA-JHU catalogues, −9999 is the missing-value sentinel of the `galSpecExtra` table (Stoughton et al. 2002; Brinchmann et al. 2004; Kauffmann et al. 2003b), so rows with unavailable components are treated as *missing data*: they are excluded from the classification and from every sSFR-based figure

[1] The embedded and predominant assignments were interchanged in parts of Tricottet et al. (2025), including Sect. 3.3. We correct the inherited catalogue labels to the definitions adopted here; isolated systems are unaffected.

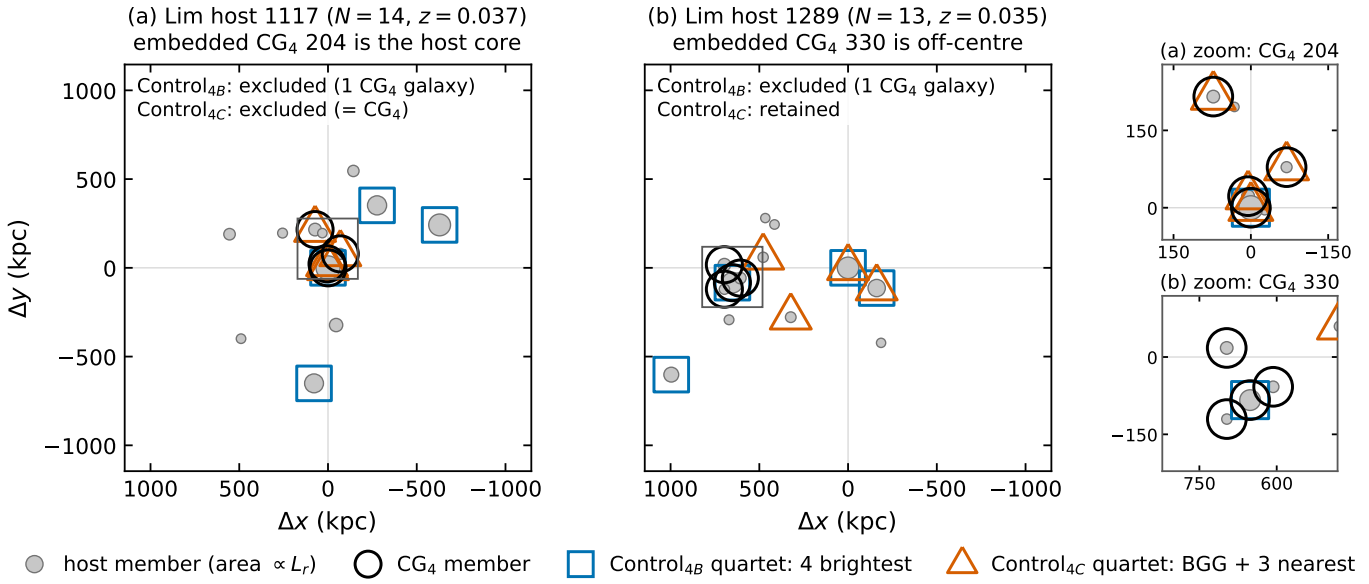


Fig. 1. Sample definitions on the sky: two Lim–Tempel groups hosting embedded compact groups, in projected physical coordinates relative to the host BGG (east to the left, common scale); disc area is proportional to r-band luminosity. Black circles mark the CG_4 members, blue squares the four brightest eligible members (the $Control_{4B}$ quartet), and orange triangles the BGG and its three nearest eligible projected companions (the $Control_{4C}$ quartet); the right-hand panels enlarge the boxed ±170 kpc windows. *(a)* CG_4 204 is an exact four-member $Control_{4C}$ core of Lim group 1117. Across the 56 non-isolated systems, 51 have any projected-core overlap and 46 are exact identities; the remainder include partial and zero overlap (Appendix C). *(b)* CG_4 330 is an off-centre substructure 670 kpc from the BGG of Lim group 1289: the $Control_{4C}$ quartet contains no compact-group galaxy and is retained, whereas the $Control_{4B}$ quartet contains the brightest CG_4 member and is excluded.

and fraction, and are reported as counts only (Table 2). All quenched and star-forming fractions are computed among classified galaxies. The missing-sSFR incidence is higher in CG_4 (7% of members, 10% of BGGs) than in the reference sample (1%). Of the 394 missing rows, 332 are BOSS-instrument spectra outside the legacy-only MPA-JHU coverage, 60 are legacy spectra whose SFR fit returned the sentinel, and 2 have no spectrum identifier (Appendix A). These provenance categories relate the missingness to catalogue and spectroscopic availability; they do not establish that availability is physically independent of star formation. Galaxies with a measured sSFR are split into quenched and star-forming classes with a two-component Gaussian mixture model (GMM) in the $\log M_\star$–$\log \mathrm{sSFR}$ plane; the boundary and the AGN handling are given in Appendix B (Fig. B.1).

Morphologies are drawn from Galaxy Zoo 1 (Lintott et al. 2008, 2011), whose debiased catalogue vote fractions (Bamford et al. 2009) are `p_el_debiased`, labelled "elliptical" in GZ1, and `p_cs_debiased`, labelled combined spiral. We define the GZ1 E class by `p_el_debiased` > 0.5, the S class by `p_cs_debiased` > 0.5, and the uncertain class when both vote fractions are finite but neither threshold is exceeded; rows with a missing vote fraction are treated as missing Galaxy Zoo data. GZ1 does not separate E from S0, so throughout this paper the E class is interpreted as a smooth early-type-like class, not as a physical sample of classical ellipticals. A cross-match with the deep-learning morphologies of Domínguez Sánchez et al. (2018) (DS18; 2436 of 3857 unique galaxies) quantifies this: of the 942 matched E-class galaxies, 92% have DS18 $T\mathrm{Type} \leq 0$, 54% are early with $P_{S0} > 0.5$, and 29% with $P_{S0} \geq 0.8$; conversely, DS18 early types with $P_{S0} > 0.5$ ($N = 730$) fall 70% in our E class and 19% in S (63% and 24% for $P_{S0} \geq 0.8$), with the E share dropping from 77% for face-on to 47% for inclined systems (DS18 $P_{\mathrm{edge\text{-}on}} < 0.1$ versus ≥ 0.5), as expected if inclined S0s expose their discs. We reserve "elliptical" for the literal GZ1 category and for external classifications that physically distinguish ellipticals. We also retrieve the MPA-JHU age-sensitive indices D_n4000 and $H\delta_A$ (Kauffmann et al. 2003b) for the stored spectra of the group galaxies, for descriptive use only (Sect. 3.5, Appendix G.3).

2.3. Galaxy sizes

Our primary size measure is the circular effective (half-light) radius of the pure Sérsic fit in the r band, $R_{e,r}$, from the structural catalogue of Simard et al. (2011): PSF-convolved GIM2D fits with re-derived sky levels and deblending designed for crowded fields, so the sizes are corrected for seeing. The catalogue is keyed to SDSS DR7 identifiers; we bridge our DR16 objects to their DR7 counterparts through the SkyServer `SpecDR7` cross-match table. Rows whose Simard redshift differs from the catalogue redshift by more than 0.005 are treated as wrong matches and dropped (9 rows). The Simard sizes, tabulated in kpc under the catalogue's own cosmology, are converted back to angular sizes with the catalogue plate scale and then to kpc with the same Planck 2015 conversion as every other projected quantity.

As a secondary, model-independent measure we use the SDSS Petrosian radii `petroR50_r` and `petroR90_r`, which also define the concentration index $C = R_{90,r}/R_{50,r}$ and provide the like-for-like bridge to Coenda et al. (2012). Pet-

rosian radii are not corrected for seeing, and the Petrosian aperture misses roughly 20% of the flux of an $n = 4$ profile, which biases $R_{50,r}$ low for concentrated galaxies (Blanton et al. 2001; Graham et al. 2005). All sizes are required to lie between 0.1 and 50 kpc after re-conversion; Sérsic fits pegged at the GIM2D bounds ($n_g \leq 0.55$ or $n_g \geq 7.9$; 60 galaxies) are excluded from the primary sample, a cut that removes 2.5% of the Simard-matched CG_4 members against 1.0–1.5% in the controls and is therefore conservative with respect to a CG_4 early-type excess; Petrosian entries must satisfy $R_{50,r} > 0$, $R_{90,r} > R_{50,r}$, and a finite positive uncertainty. The completeness of both measures is given in Appendix G.3.

2.4. Statistical approach

Our main question is whether compact-group galaxies retain distinct properties once differences in stellar mass, redshift, galaxy rank, and group properties are taken into account. The three controls address physically different comparisons, so we compare CG_4 with each one separately. Galaxy-level logistic models quantify the adjusted E-versus-S and quenched-versus-star-forming contrasts; group-level matching compares the satellite mix of whole quartets; and within-host models compare compact-group members with their non-compact co-members. The covariates are stellar mass, redshift, rank, group luminosity, and the quartet velocity dispersion σ_v; because σ_v from four tracers is noisy, every analysis that uses it is repeated without it (Appendix H).

The independence unit is the *physical* Lim group: standard errors are clustered by physical group and resampling is blocked by physical group or matched pair, because the four galaxies of one group are not independent. We report two-sided p-values with effect sizes and confidence intervals for the primary morphology, star-formation, matching, host, and size analyses, and reserve multiplicity control for the exploratory batteries. The p-value conventions, the covariate-completeness rule, the overlap coefficient, and the treatment of the overlapping control samples in pooled summaries are collected in Appendix D.

3. Results

3.1. Morphology

CG_4 galaxies are more often in the GZ1 E class than the galaxies of every control. Tables 2 and 3 give the raw class counts; among galaxies with a usable E or S classification, the CG_4 E-class fraction is 59% against 47%, 48%, and 32% in $Control_{4B}$, $Control_{4C}$, and RG_4 (Fisher exact tests: $p = 6.8 \times 10^{-4}$, $p = 1.9 \times 10^{-3}$, and $p < 10^{-4}$). The 38 CG_4 rows with uncertain or missing Galaxy Zoo classifications are excluded rather than assigned to either class. Satellites account for most of the difference: among usable BGG classifications the E-class fraction is 67.9% in CG_4 and 44.7% in RG_4 (Fisher $p = 1.00$, 1.00, and 0.03 against $Control_{4B}$, $Control_{4C}$, and RG_4), and restricted to star-forming galaxies the E/S mix is compatible with every control (two-sided Fisher exact test: $p = 0.332$, $p = 0.468$, $p = 0.226$), so star-forming E-class galaxies do not produce the raw excess.

The mass-binned view separates BGGs from satellites and shows that the E-class excess is primarily a satellite result rather than a stellar-mass artefact (Fig. 2). The quenched fractions do not show an analogous CG_4-specific pattern.

The excess survives adjustment for stellar mass, redshift, rank, group luminosity, and σ_v (Table 4). It is clearest against $Control_{4B}$ and RG_4 and weakest against the projected cores $Control_{4C}$, without implying that the three estimates differ significantly from one another; omitting the noisy σ_v covariate leaves the interpretation unchanged (Appendix H).

In the pooled model over the deduplicated control catalogues (Appendix G), the satellite E-class odds ratio is 2.08 (95% CI [1.51, 2.88]) ($p < 10^{-4}$), whereas the BGG estimate is 0.78 (95% CI [0.39, 1.53]) ($p = 0.464$); the pooled all-galaxy odds ratio is 1.77 ($p = 1.4 \times 10^{-4}$).

Whole compact groups contain more E-class satellites than matched ordinary groups. Treating each quartet as the unit, we count its E-class satellites ($N_{\mathrm{E,sat}} \in \{0, 1, 2, 3\}$ among classified satellites) and match CG_4 groups one-to-one to controls on redshift, BGG stellar mass, total quartet luminosity, and σ_v (propensity caliper 0.2 SD of the logit), separately for each control (Table 5). The CG_4-minus-control difference in E-class satellite fraction is $\Delta = 0.16$ against $Control_{4B}$ (95% CI [0.04, 0.28]; host-component sign-flip $p = 0.013$) and 0.20 against RG_4 ($p = 0.033$), while against the projected cores $Control_{4C}$ it has the same sign but is not significant ($\Delta = 0.11$, [−0.01, 0.23], $p = 0.083$). Adding the mean satellite stellar mass to the match preserves the ordering ($\Delta = 0.20$, 0.09, and 0.26). Without σ_v in the match the $Control_{4B}$ difference grows to 0.28 because a different set of control quartets, from dynamically cooler hosts, is selected (only 7 of 54 control groups are shared), while the $Control_{4C}$ and RG_4 values change little (Appendix H).

Individually matched galaxies confirm the shift. Matching 234 of the 248 CG_4 galaxies one-to-one to ordinary-group galaxies of the deduplicated control pool on stellar mass, redshift, rank, group luminosity, and σ_v (exact rank strata, common support, 0.2-SD caliper; maximum absolute standardized mean difference 0.08 after matching, Appendix G), the matched E-class fraction difference is 0.14 (95% block-bootstrap interval [0.04, 0.24]; $p = 0.007$), rising to 0.19 for the 130 satellite pairs alone ($p = 0.004$).

Within their own host groups, the CG_4 association is substantially attenuated after accounting for host-centric radius, although the confidence interval remains compatible with a non-negligible residual association. For the 19 embedded and 37 predominant systems, whose quartets live inside richer Lim groups (54 distinct hosts; 958 member galaxies, 224 of them in a CG_4), we compare the CG_4 members with the non-CG_4 members of the same host by conditional logistic regression stratified by host, which removes every host-level property by design, with stellar mass, host luminosity rank, and projected host-centric radius as covariates. The E-class odds ratio for CG_4 membership is 1.47 (95% CI [0.92, 2.35]; $p = 0.108$; 791 galaxies in 53 informative strata), and rises to 1.90 ([1.23, 2.95]; $p = 0.004$) when host-centric radius is omitted. Radius therefore helps locate where the observational association resides; this coefficient change is neither a causal mediation fraction nor proof of environmental equivalence. A fixed-effects GLM gives the same direction (Appendix H).

Table 2. sSFR-class counts

Sample	Quenched	Star-forming	No sSFR
CG_4	146 (63%)	84 (37%)	18 (7%)
BGG	45 (80%)	11 (20%)	6 (10%)
Satellites	101 (58%)	73 (42%)	12 (6%)
$Control_{4B}$	1593 (61%)	1019 (39%)	180 (6%)
BGG	533 (83%)	107 (17%)	58 (8%)
Satellites	1060 (54%)	912 (46%)	122 (6%)
$Control_{4C}$	1560 (60%)	1061 (40%)	191 (7%)
BGG	537 (83%)	107 (17%)	59 (8%)
Satellites	1023 (52%)	954 (48%)	132 (6%)
RG_4	101 (46%)	118 (54%)	5 (2%)
BGG	38 (72%)	15 (28%)	3 (5%)
Satellites	63 (38%)	103 (62%)	2 (1%)
SDSS reference	7705 (15%)	44192 (85%)	634 (1%)

Notes. Quenched and star-forming percentages use classified galaxies as the denominator; No-sSFR percentages use all galaxies in the row. The SDSS non-AGN sample is a deeper orientation reference, not a control sample and enters no test.

Table 3. Morphology-class counts

Sample	**GZ1 E**	**GZ1 S**	**Unc.**	**No GZ**
CG_4	124 (50%)	86 (35%)	7 (3%)	31 (12%)
$Control_{4B}$	1106 (40%)	1266 (45%)	116 (4%)	304 (11%)
$Control_{4C}$	1118 (40%)	1229 (44%)	144 (5%)	321 (11%)
RG_4	60 (27%)	129 (58%)	15 (7%)	20 (9%)
SDSS ref.	*10091 (19%)*	*34715 (66%)*	*7725 (15%)*	*0 (0%)*

Notes. Percentages use each sample's final row count. Inferential E/S comparisons condition on the usable E/S subset. "No GZ" denotes missing vote fractions and is distinct from finite-vote uncertain classifications. The SDSS reference is for orientation only.

Table 4. Adjusted per-control models

Outcome	Population	Control	N (hosts)	CG_4 odds ratio (95% CI)	p
E versus S	All galaxies	$Control_{4B}$	2582 (738)	2.14 [1.58, 2.90]	$< 10^{-4}$
		$Control_{4C}$	2557 (740)	1.67 [1.25, 2.24]	5.9×10^{-4}
		RG_4	399 (113)	2.81 [1.79, 4.38]	$< 10^{-4}$
E versus S	Satellites	$Control_{4B}$	1929 (735)	2.63 [1.88, 3.68]	$< 10^{-4}$
		$Control_{4C}$	1901 (734)	1.93 [1.40, 2.66]	$< 10^{-4}$
		RG_4	299 (113)	3.03 [1.72, 5.34]	1.3×10^{-4}
Quenched versus star-forming	All galaxies	$Control_{4B}$	2842 (755)	1.80 [1.26, 2.58]	0.001
		$Control_{4C}$	2851 (758)	1.25 [0.91, 1.71]	0.173
		RG_4	449 (115)	1.40 [0.80, 2.47]	0.238
Quenched versus star-forming	Satellites	$Control_{4B}$	2146 (755)	2.16 [1.51, 3.09]	$< 10^{-4}$
		$Control_{4C}$	2151 (757)	1.38 [1.00, 1.90]	0.052
		RG_4	340 (115)	1.85 [0.99, 3.46]	0.052
Quenched versus star-forming	Satellites; + GZ1 E/S	$Control_{4B}$	1914 (732)	1.85 [1.13, 3.02]	0.015
		$Control_{4C}$	1890 (732)	1.29 [0.83, 2.01]	0.259
		RG_4	296 (112)	1.39 [0.65, 2.97]	0.393

Notes. Separate cluster-robust logistic models compare CG_4 with each control. All models include stellar mass, redshift, galaxy rank, group luminosity, and velocity dispersion; confidence intervals and p-values are Wald quantities clustered by physical Lim group. The final three satellite-quenching rows additionally include the conservative binary GZ1 E-versus-S indicator and therefore exclude uncertain or missing morphology.

3.2. Star formation at fixed morphology

The star-formation contrast is weaker and control-dependent (Table 4). It is clearest against RG_4 in the raw data, but the adjusted and matched results do not support a control-independent excess of quenching; assigning all missing-sSFR galaxies to either extreme class does not erase the raw RG_4 contrast (Appendix B).

Morphology accounts for part of the satellite-quenching contrast, while the remaining association is control-dependent. In direct satellite-only models that add the binary GZ1 E/S indicator to stellar mass, redshift, group luminosity, and velocity dispersion, the CG_4 odds ratios are 1.85 (95% CI [1.13, 3.02]; $p = 0.015$), 1.29 ([0.83, 2.01]; $p = 0.259$), and 1.39 ([0.65, 2.97]; $p = 0.393$) against $Control_{4B}$, $Control_{4C}$, and RG_4. Thus a residual associa-

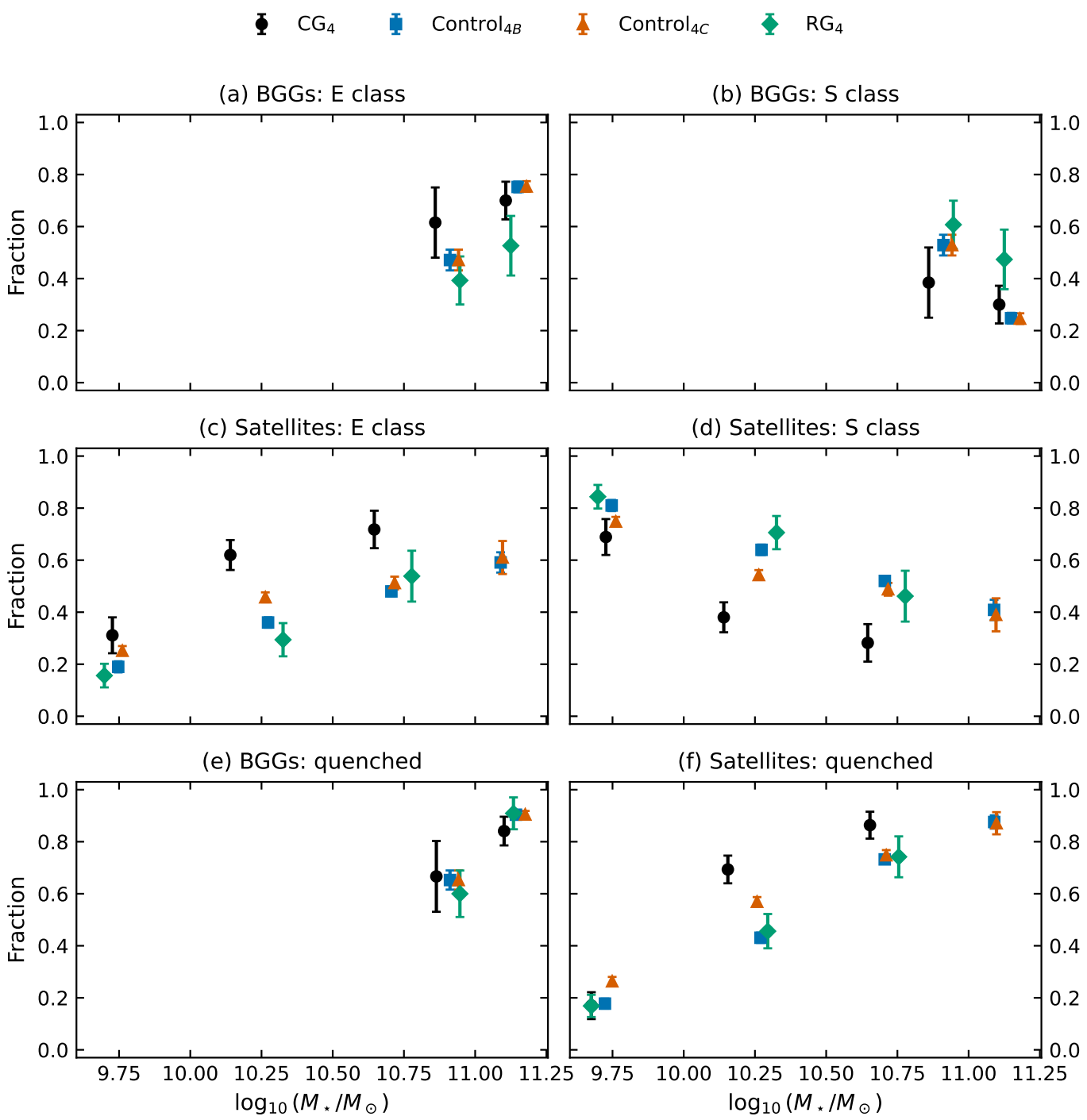


Fig. 2. Morphology and quenched fractions versus stellar mass. Rows show BGG E/S fractions, satellite E/S fractions, and quenched BGG/satellite fractions. All six panels use edges 7.0, 10.0, 10.5, 11.0, 12.5; points are placed at the median mass in each bin. Binomial uncertainties are drawn with G. Mamon's `binomialerrorplot`; empty/full bins use its one-sided Wilson limits. Bins with fewer than five galaxies or three physical groups are omitted.

tion remains against $\mathrm{Control}_{4B}$ but is not recovered clearly against $\mathrm{Control}_{4C}$ or RG_4; this comparator dependence does not identify a causal quenching mechanism. On the exact $\mathrm{Control}_{4B}$ morphology-complete sample, omitting morphology raises the odds ratio from 1.85 to 2.67, so the attenuation is not caused by changing the fitted population.

The descriptive conditional fractions are $P(\mathrm{Q}|\mathrm{E}) = 0.908$, 0.903, 0.887, and 0.850 for CG_4, $\mathrm{Control}_{4B}$, $\mathrm{Control}_{4C}$, and RG_4 (Table B.2). Within the common morphology-available and sSFR-classified population used by the Kitagawa decomposition, each total CG_4-minus-control difference is split exactly into a morphology-composition term and a conditional-within-morphology term (Appendix B, Table B.3). The host-blocked 95% intervals for the conditional terms are $[-0.073, +0.039]$, $[-0.056, +0.056]$, and $[-0.035, +0.136]$; these compatible ranges do not demonstrate equality. The individually matched quenched-fraction difference is 0.03 (95% CI

Table 5. Matched satellite-composition contrasts

Control	Pairs	Δ	95% CI	$p_{\rm perm}$
$\mathrm{Control}_{4B}$	54	0.160	[0.042, 0.280]	0.013
$\mathrm{Control}_{4C}$	54	0.114	[−0.012, 0.233]	0.083
RG_4	28	0.196	[0.036, 0.351]	0.033
Pooled	54	0.167	[0.039, 0.290]	0.013

Notes. Δ is the matched mean E-class satellite-fraction difference (CG_4 minus control), with 95% bootstrap intervals from 9999 resamples of connected components in the bipartite treated-quartet/control-host graph; $p_{\rm perm}$ flips signs at the same component level. The final row uses the deduplicated control pool.

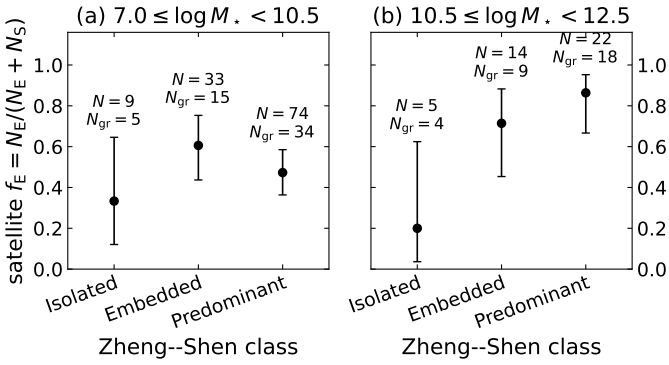


Fig. 3. Mass-binned satellite E-class fractions. Panels use the common broad mass bins shown above each axis; points and bars give class fractions and descriptive Wilson 95% intervals, with classified-galaxy and contributing-group counts.

[−0.06, 0.12]; $p = 0.526$). With the existing fixed-threshold diagnostic, $\log_{10}(\mathrm{sSFR/yr^{-1}}) = -11.0$, the CG_4 and RG_4 satellite star-forming fractions are 41.4% and 60.2% (difference −18.8 points; $p = 5.3 \times 10^{-4}$).

No robust CG_4 displacement of the star-forming main sequence is established: the unmatched median offsets are comparison-dependent, while the matched star-forming pairs give a small mean difference whose interval includes zero. The sequence fits are described in Appendix E; the cumulative distributions and full numerical comparison are retained in the online supplement. The quenched-sequence residuals are also shown there, but are weakly informative because MPA-JHU sSFRs of quenched galaxies are largely D_n4000-calibrated.

3.3. Compact-group classes

The early-type-like excess is concentrated among the compact groups that live inside richer hosts. Figure 3 compares the satellite E-class fraction in two broad stellar-mass bins; the aggregate fractions $f_{\rm E} = N_{\rm E}/(N_{\rm E} + N_{\rm S})$ for the three Zheng–Shen classes of Sect. 2.1: $f_{\rm E} = 0.638^{+0.122}_{-0.143}$ for the 19 embedded and $0.562^{+0.095}_{-0.100}$ for the 37 predominant systems, against $0.286^{+0.261}_{-0.169}$ for the 6 isolated ones (aggregate Wilson 95% intervals). The isolated compact groups are compatible with the loose RG_4 groups (group-level permutation test on the per-group mean satellite $f_{\rm E}$, $p = 1.00$) and sit below the rest of CG_4 ($p = 0.023$), whereas the two non-isolated classes do not differ detectably (predominant-versus-embedded odds ratio at fixed stellar mass 0.97, $p = 0.94$). The host halo masses differ strongly between the classes (median $\log M_{200c} = 12.86$, 13.83, and 13.13 for isolated, embedded, and predominant systems), yet adding $\log M_{200c}$ to the class model changes neither the class coefficients nor the conclusion ($\log M_{200c}$ odds ratio 0.80 per standard deviation, $p = 0.31$). With only 6 isolated groups, this class contrast is exploratory; other group-scale descriptors are examined in Appendix F.

3.4. Local projected density

Including local projected density attenuates the CG_4 coefficient, but the amount is not well determined. For each of the 6076 catalogue rows we compute $T_i = \sum_j M_{\star,j}/R_{ij}^3$ over selected quartet co-members (R_{ij} in kpc; unmeasured companion masses contribute zero, making T_i a lower bound). For the pooled sensitivity fit, controls are deduplicated by objid as in Appendix G, retaining the quartet assignment in the priority order RG_4, $\mathrm{Control}_{4B}$, $\mathrm{Control}_{4C}$. The E-class model adjusts for stellar mass and satellite status, with physical-group clustering. With the index, the CG_4 odds ratio is 1.27 (95% CI [0.94, 1.71]; $p = 0.12$; 3260 galaxies). Retaining all catalogue rows instead gives an attenuation from 1.95 to 1.39 ($p = 0.031$), with an index odds ratio of 1.42 per 1.07-dex standard deviation ($p < 10^{-4}$); this check weights multiply selected controls more heavily. On that catalogue-row frame, the $\log T_i$ overlap coefficient is 0.54 and the median gap 1.1 dex; restricting to common support gives OR 1.22 ($p = 0.20$). Conditioning on an index built from the separations defining compact groups partly conditions on selection, so attenuation does not establish mediation. The index traces projected neighbours, not tidal history; the host-inclusive check is given in Appendix G.3.

3.5. Other galaxy properties

None of the other galaxy properties shows an offset as robust as the morphology excess (Fig. 4). Strong Hα emission (Hα EW ≤ −3 Å, emission negative) is rarer in CG_4: 70 of 217 members with measurements (32%), against 41%, 43%, and 57% in $\mathrm{Control}_{4B}$, $\mathrm{Control}_{4C}$, and RG_4 (Fisher $p = 0.012$, $p = 0.002$, $p < 10^{-4}$). The deficit is not fully explained by the different morphology mix: after adjustment for stellar mass and GZ1 E/S class, the CG_4 odds ratios for strong Hα emission are 0.55, 0.65, and 0.56 against $\mathrm{Control}_{4B}$, $\mathrm{Control}_{4C}$, and RG_4, respectively, with the first two contrasts statistically clear. The AGN-like BPT line-ratio fraction (above the Kewley et al. 2001 line, or flagged as AGN without line ratios; composites between the Kauffmann et al. 2003a and Kewley et al. 2001 lines) is 38% in CG_4, and its mass- and environment-adjusted CG_4 odds ratio is 1.17 ($p = 0.364$); this diagnostic does not measure all nuclear activity. The D_n4000 index follows stellar mass in every sample; the CG_4 medians sit above the controls only in the 10.0–10.5 bin ($D_n4000 = 1.77$ against 1.59, 1.68, and 1.56), where the CG_4 E-class fraction also peaks relative to the controls, and coincide with them elsewhere; a post-starburst classification from D_n4000 and Hδ_A is deferred to a follow-up paper.

No clear adjusted size offset is detected between CG_4 and ordinary-group galaxies. With the seeing-corrected Simard radii, the satellite size offset at fixed stellar mass, redshift, and group-scale covariates is −0.010 dex (−2%; 95% CI [−0.039, 0.019]; $p = 0.507$; 2575 satellites in 754

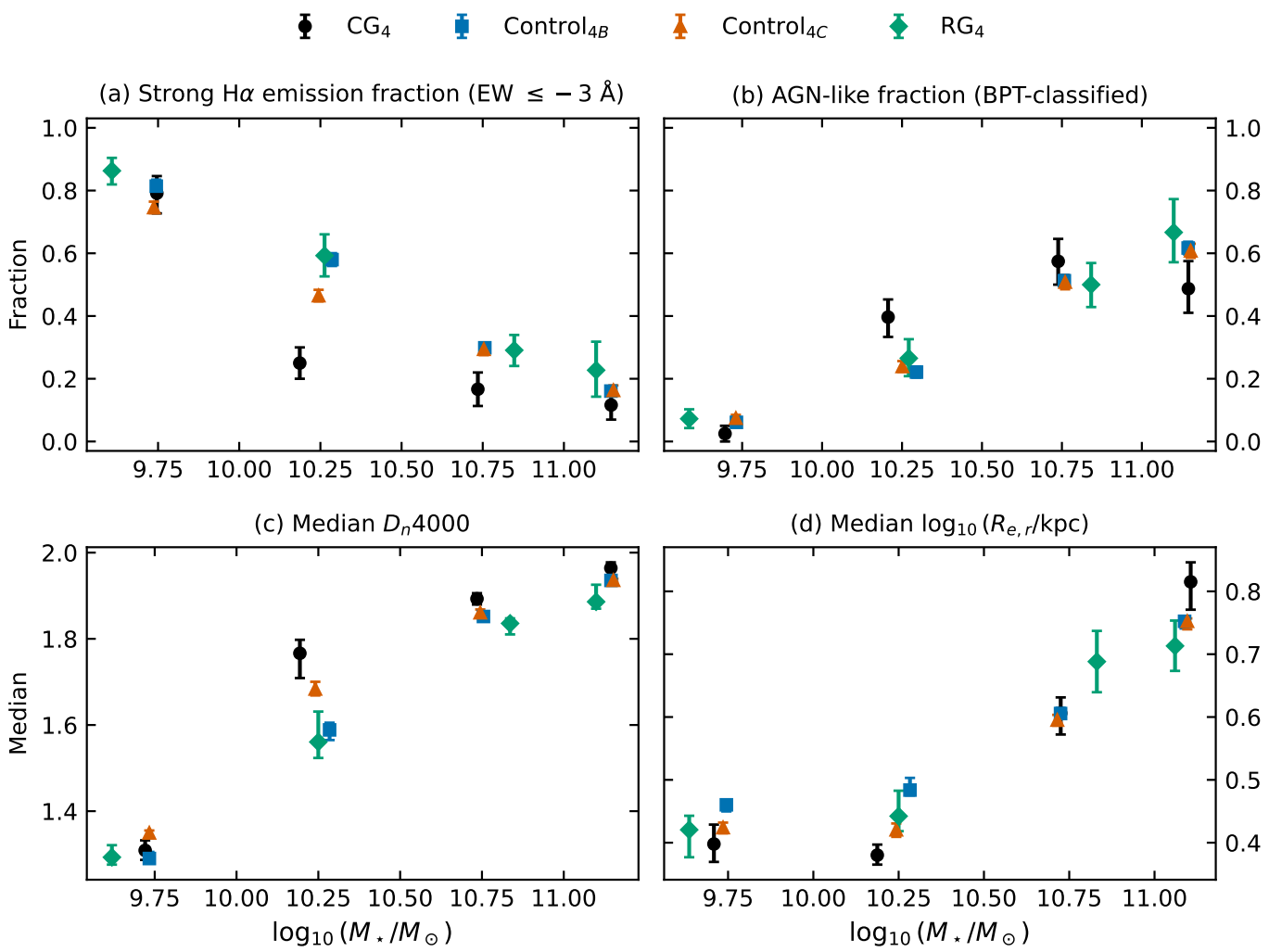


Fig. 4. Other galaxy properties versus stellar mass: (a) strong-Hα fraction; (b) AGN-like fraction among BPT-classified galaxies; (c) median D_n4000; (d) median seeing-corrected effective radius $\log_{10}(R_{e,r}/\mathrm{kpc})$. Points use the mass bins of Fig. 2; bars here are 16th–84th percentile intervals from a bootstrap blocked by physical group. These comparisons are descriptive.

physical groups), and −0.006 dex for all galaxies ($p = 0.653$); the per-control estimates are −0.031 dex against $\mathrm{Control}_{4B}$, −0.001 dex against $\mathrm{Control}_{4C}$, −0.021 dex against RG_4 (Appendix I; see the online supplement). The one-to-one matched difference is −0.019 dex for the 166 pairs with Simard sizes (95% CI $[-0.064, 0.026]$; $p = 0.400$), and −0.033 dex for the 234 pairs with uncorrected Petrosian radii ($p = 0.030$). The Simard measure resolves 77% of the CG_4 galaxies against 71% of the pooled controls, so the Simard coefficients carry a differential-completeness caveat that the Petrosian rerun does not share (Petrosian satellite coefficient −0.019 dex, $p = 0.111$). Colours, magnitude gaps, and projected phase-space trends are likewise control-dependent or weak; they are reported in Appendix F.

3.6. Robustness

The morphology excess does not depend on the vote threshold, on the classifier, or on the crowded closest pairs (Table 6). Excluding every galaxy whose nearest catalogued neighbour lies within 55 arcsec, the SDSS fibre-collision scale, removes 19% of CG_4 members against 6–10% of the controls and moves the adjusted E-class odds ratios from 2.14 to 2.16 ($\mathrm{Control}_{4B}$), 1.67 to 1.71 ($\mathrm{Control}_{4C}$), and 2.81 to 2.71 (RG_4). Across classification thresholds of 0.4–0.8 the pooled adjusted odds ratio stays between 1.73 and 2.19, the continuous debiased E vote fraction p_E itself is offset by 0.089 ($p < 10^{-4}$), and the Galaxy-Zoo-independent Sérsic index ($n_g > 2.5$; 75% coverage) gives an odds ratio of 1.60 ($p = 0.002$). With the DS18 $T\mathrm{Type} \leq 0$ classification as the outcome, the per-control CG_4 odds ratios are 2.11, 1.49, and 2.22 (all $p \leq 0.017$; $n = 1822, 1913, 324$). The crowding, threshold, and σ_v reruns are detailed in Appendix H.

4. Interpretation and comparison with previous work

4.1. An early-type-like satellite excess linked to dense group cores

Compact groups of four galaxies host more early-type-like satellites than any of the three ordinary-group comparison samples. The three adjusted contrasts are ordered in the same sense as the characteristic compactness of the controls, with the smallest contrast against the projected

Table 6. Morphology-contrast robustness

Test	Result
55-arcsec neighbour exclusion	all-row E: 52.5% / 40.1% (CG_4/ $Control_{4C}$); E/S Fisher $p = 0.002$
Add close-neighbour flag to model	CG_4 OR 1.76 [1.30, 2.39]; $p = 2.6 \times 10^{-4}$
DS18 TType ≤ 0	OR 2.11 / 1.49 / 2.22 ($Control_{4B}$ / $Control_{4C}$ / RG_4); all $p \leq 0.017$
Threshold sweep 0.4–0.8	OR 1.73–2.19
Continuous p_E offset	0.089 [0.047, 0.131]; $p < 10^{-4}$
Sérsic $n_g > 2.5$	OR 1.60; $p = 0.002$
Without σ_v	OR 2.22 / 1.68 / 2.76 ($Control_{4B}$ / $Control_{4C}$ / RG_4)

Notes. Odds ratios are adjusted CG_4 coefficients for E-class membership; the continuous row is the adjusted offset in debiased E vote fraction p_E. P-values are unadjusted. Rows are separated by robustness family.

cores $Control_{4C}$; this is a descriptive ordering, not a formal between-control trend (Sect. 3.1). The CG_4 association is substantially attenuated after accounting for host-centric radius, although the confidence interval remains compatible with a non-negligible residual association. Montaguth et al. (2026b) already examined quenched and early-type fractions with host-centric position and local density around non-isolated compact groups. Our narrower addition combines conditional shared-host modelling and individual covariates with the explicit quartet-control analyses; differences in samples and morphology definitions make the results complementary. Restricting the individually matched comparison to satellites raises the E-class fraction difference to 0.19. Present-day stellar mass may itself encode part of a galaxy's environmental history, so fixed-mass estimates describe the remaining association rather than the total accumulated transformation.

The BGG–satellite separation matters when comparing with Martínez et al. (2013), whose brightest-galaxy selections involved group luminosity, morphology or concentration, and magnitude gap. Our lack of a consistent adjusted BGG excess need not conflict with their result given the different samples, morphology measures, and adjustments. The luminous-BGG selection concentrates our samples at high stellar mass, where most BGGs are already quenched (Table 2), potentially limiting the observable environmental contrast.

The 46 exact host-core identities among 56 non-isolated CG_4 systems (Appendix C) connect this result to the within-sample compactness trends: at fixed stellar mass, smaller median pairwise separations accompany higher satellite E-class probabilities within CG_4, $Control_{4B}$, and $Control_{4C}$ separately (Table F.1). These results are consistent with many non-isolated CG_4 systems sampling the dense end of a broader group-core compactness–morphology relation, but do not test common slopes or intercepts. Local projected-density adjustment also attenuates the CG_4 association (Sect. 3.4), without identifying causal ordering. This interpretation accords with the structural signatures in recent S-PLUS studies (Montaguth et al. 2023, 2025, 2026a,b), while allowing other host properties, selection effects, and evolutionary pathways to contribute. The 6 isolated systems have lower satellite E-class fractions than the rest of CG_4 but cannot establish a distinct evolutionary stage (Sect. 3.3). Host-mass adjustment leaves the class contrasts unchanged (Appendix G.3); this does not exclude a host contribution.

Stellar-population diagnostics could in principle separate an accumulated enrichment history from a shorter star-formation timescale, but a metallicity analysis requires its own treatment of selection, aperture coverage, and radial gradients. The SDSS fibres sample central regions whose physical extent varies with redshift, complicating the interpretation of age and metallicity gradients, so that analysis is outside the present scope.

4.2. GZ1 E-class morphologies and transition galaxies

Our E class is a smooth early-type-like proxy with a substantial S0 content (Sect. 2.2). Galaxy Zoo 1 cannot determine whether the excess comprises S0s, ellipticals, or both; we reserve that question for a dedicated follow-up analysis of the DS18 classes by environment. This caveat matters when comparing with the S-PLUS studies of compact-group transition galaxies and mass–size trends (Montaguth et al. 2023, 2025, 2026a,b): those studies use hybrid structural–colour classifications, identify transition populations, and adopt their own size definitions, whereas we use GZ1 E/S classes and seeing-corrected Simard radii. Their result therefore does not predict a universal size deficit for every CG_4 GZ1 E-class galaxy. Our three control samples instead test which apparent CG_4 signatures survive across genuine four-member, luminosity-selected, and BGG-centred projected-core quartets.

4.3. Why the star-formation and size signals are weaker

Morphology accounts for part of the sSFR-defined quenching contrast; the remaining adjusted association persists against $Control_{4B}$ but is not recovered clearly against $Control_{4C}$ or RG_4 (Sect. 3.2). The decomposition's conditional intervals span zero without demonstrating equality or superseding the regressions. The strong-Hα deficit persists at fixed broad morphology and stellar mass. Its EW measures line emission relative to stellar continuum within the 3-arcsec legacy SDSS fibre and can include non-star-forming excitation from AGN, shocks, or old stars. MPA-JHU total sSFRs instead incorporate photometric aperture corrections and statistical estimates for weak-line systems (Brinchmann et al. 2004; Salim et al. 2007). These measurement differences may contribute to the differing contrasts without establishing central quenching. Spectroscopic incompleteness also matters: 19% of catalogued CG_4 galaxies have a neighbour below the 55-arcsec fibre-collision scale, against 8% of pooled controls (10% in $Control_{4C}$). Missing companions could hide brief starbursts or post-starburst phases and bias the tidal index downward; aperture corrections may also be less secure for interacting or blended members. The comparison-dependent sSFR and main-sequence estimates therefore remain uncertain.

The size comparison is sensitive to methodology. Coenda et al. (2012) reported a tendency towards smaller compact-group galaxies at fixed luminosity with Petrosian

radii within concentration-selected morphological subsamples, although their more restrictive $C > 3$ selection showed no clear environmental trend. Our seeing-corrected Simard radii give no clear fixed-mass offset, with a 95% interval still permitting a deficit of roughly 0.04 dex. The matched Simard and Petrosian differences also use unequal available-pair subsets (Sect. 3.5). The larger Simard-minus-Petrosian offsets for E-class galaxies (Appendix I) make morphology mix relevant, without measuring its contribution to the environmental contrast. Profile-dependent size estimates offer a possible methodological contribution to differences between studies, but sample selection, luminosity-versus-mass conditioning, and structural classification also differ. Modest intrinsic size differences remain allowed.

5. Conclusion

We have tested whether galaxies in CG_4 compact groups remain special after removing split systems and comparing them with RG_4, $Control_{4B}$, and $Control_{4C}$. The conclusions apply to low-redshift, luminous, exactly four-member non-split HMCGs present in the required parent catalogues and satisfying the BGG and group-luminosity constraints. Morphology-based results additionally apply only where GZ1 classifications are available, and the sample is not universally stellar-mass complete for satellites. $Control_{4C}$ is a BGG-centred projected-core comparator rather than a compactness-matched sample, so the coefficients do not isolate an "effect of compactness alone."

The main result is an early-type-like satellite excess. CG_4 galaxies are more often in the GZ1 E class than all three regular-group comparisons after accounting for galaxy and group properties; the difference is strongest against $Control_{4B}$ and RG_4 and weakest against the projected cores $Control_{4C}$. BGGs show no consistent adjusted excess across the controls, and within their hosts the residual CG_4 excess is no longer statistically clear once central location is held fixed. A projected tidal-density proxy attenuates the CG_4 coefficient, but its entanglement with compact-group selection and imperfect common support preclude a causal interpretation.

Morphology accounts for part of the satellite-quenching contrast. After simultaneous adjustment for morphology and the principal covariates, a residual association remains against $Control_{4B}$ but is not recovered clearly against $Control_{4C}$ or RG_4; this is control-dependent evidence, not a causal mechanism. Star-forming CG_4 galaxies show no robust main-sequence offset, and the secondary strong-Hα deficit persists after adjustment for broad morphology and stellar mass. At fixed stellar mass the seeing-corrected satellite size estimate is -0.010 dex with a 95% interval spanning zero, so no clear adjusted size offset is detected. Colours, AGN-like line ratios, magnitude gaps, D_n4000, and projected phase-space trends identify no simple alternative driver.

We therefore interpret this selected luminous, low-redshift, four-member, non-split catalogue population as shifted towards early-type-like satellite morphologies. The association's attenuation with host-centric radius locates it observationally within the host but does not identify a unique mechanism. The isolated subset is too small to establish a distinct mode. Explicit E–S0–spiral classifications, bulge–disc structure, and stellar-population diagnostics will be needed to distinguish possible pathways.

Data availability

The input data are public SDSS, MPA-JHU, Galaxy Zoo, Simard et al. (2011), group, and compact-group catalogues cited above. Derived galaxy and group tables will be made available through CDS at publication. The analysis and paper-generation code is available at `https://github.com/MatthieuTricottet/CG_Gals`; the full SDSS query used for the reference sample is in `src/data_loader.py`, the DR12 spectrum-provenance and spectral-index retrievals in `src/ssfr_quality_audit.py` and `src/spectral_indices.py`, and the read-only diagnostics of this revision under `results/diagnostics/`. The article release, including online diagnostic supplements, will be archived with a DOI at acceptance.

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Appendix A: SDSS data query

For the SDSS reference sample, we extract `sfr_tot_p50`, `specsfr_tot_p50`, and `lgm_tot_p50` from `galSpecExtra`, `h_alpha_eqw` and the [N II], Hα, [O III], and Hβ fluxes from `galSpecLine`, and `p_el_debiased` and `p_cs_debiased` from `zooSpec`. In that query, these spectroscopic tables are joined to `SpecObj` by `specObjID`, while the photometric object is linked through `SpecObj.bestObjID=PhotoObj.objID`. The full result is split into the AGN-inclusive table and the non-AGN reference used for GMM calibration. The group samples instead inherit their DR12 `sfr_tot_p50`, `lgm_tot_p50`, and stored `specObjID` from the parent catalogues; the Galaxy Zoo vote fractions and the emission-line quantities are added to those rows by photometric `objid`, and the `galSpecIndx` indices `d4000_n` and `lick_hd_a` (with errors; sentinel values and non-positive errors treated as missing) by the stored DR12 `specObjID`. Their final log sSFR is the stored difference `sfr_tot_p50−lgm_tot_p50`.

We audited all 6076 final sample rows against `SpecObjAll` in DR12, retaining the stored `specObjID` even when the same photometric `objid` has several spectra (looking these DR12 identifiers up in DR16 fails for reprocessed BOSS observations, whose identifier encodes the reduction version). Of the 394 rows without a valid sSFR, 332 are BOSS-instrument spectra (`survey` = `boss`, `run2d` = `v5_7_0`) without any `galSpecExtra` or `galSpecInfo` row, because the MPA-JHU tables cover the legacy `run2d` = 26 reductions only; 60 are legacy spectra with a `galSpecExtra` row, a valid stellar mass, `reliable` = 1, and `zWarning` = 0, whose `sfr_tot_p50` is the −9999 sentinel; and 2 (both in Control$_{4B}$) have no stored spectrum identifier. Per sample the BOSS/sentinel counts are 14/4 in CG$_4$, 149/29 in Control$_{4B}$, 164/27 in Control$_{4C}$, and 5/0 in RG$_4$. The missing rows are not low-signal-to-noise spectra: their median `snMedian_r` is 45 against 39 for the classified rows (rank-sum $p < 10^{-4}$ over unique spectra), and the same holds within each sample. The missing class is therefore a coverage gap of the MPA-JHU value-added catalogues plus a small number of failed SFR fits, unrelated to the star-formation properties of the galaxies; its extreme-assignment bounds are given in Appendix B.

Appendix B: sSFR classification details

We identify BPT-classifiable galaxies by requiring measured values of both emission-line ratios

$$\log_{10}([\mathrm{N\,II}]\lambda6584/\mathrm{H}\alpha) \quad \text{and} \quad \log_{10}([\mathrm{O\,III}]\lambda5007/\mathrm{H}\beta).$$

Among these galaxies, we flag AGN-like systems if they satisfy either

$$\log_{10}\frac{[\mathrm{N\,II}]}{\mathrm{H}\alpha} > 0,$$

or if they lie above the empirical demarcation of Kauffmann et al. (2003a):

$$\log_{10}\frac{[\mathrm{O\,III}]}{\mathrm{H}\beta} > \frac{0.61}{\log_{10}([\mathrm{N\,II}]/\mathrm{H}\alpha) - 0.05} + 1.3,$$

in which case they are excluded from the GMM calibration sample. This calibration flag is deliberately more inclusive than the three-class BPT scheme of Sect. 3.5, in which composite galaxies lie between the Kauffmann et al. (2003a) and Kewley et al. (2001) demarcations and only galaxies above the Kewley et al. (2001) line are labelled AGN-like, because its sole purpose is to keep possible AGN contamination out of the sSFR calibration. Galaxies with missing diagnostic line ratios are BPT-unclassified, not proven non-AGN; they are retained for the sSFR status assignment when valid SDSS stellar mass and sSFR estimates are available, while the AGN-like analysis treats them separately. Reference-sample rows with unavailable `specsfr_tot_p50`, and group-sample rows with unavailable `sfr_tot_p50` or `lgm_tot_p50`, including the −9999 sentinel, have no valid sSFR estimate; they are treated as missing data throughout: excluded from the GMM calibration where applicable, never assigned an sSFR class, omitted from every sSFR figure, and reported as counts only (Sect. 2.2). Galaxies with a measured sSFR are divided into quenched and star-forming classes with a two-component Gaussian Mixture Model (GMM) in the $\log M_\star$-log sSFR plane (McLachlan & Peel 2000); the boundary is the locus where the two component posterior probabilities are equal, and the resulting classification for the SDSS non-AGN reference is shown in Fig. B.1. A fixed-threshold check confirms that the qualitative hierarchy is not tied to the exact GMM split (Sect. 3.2).

Table B.1. Missing-sSFR extreme-assignment bounds

Scope	Control	Missing (%) CG$_4$	ctrl	$\Delta f_Q^{\mathrm{obs}}$	min	max
All	Control$_{4B}$	7.3	6.4	0.025	−0.046	0.091
All	Control$_{4C}$	7.3	6.8	0.040	−0.034	0.107
All	RG$_4$	7.3	2.2	0.174	0.115	0.210
Sat.	Control$_{4B}$	6.5	5.8	0.043	−0.021	0.101
Sat.	Control$_{4C}$	6.5	6.3	0.063	−0.005	0.122
Sat.	RG$_4$	6.5	1.2	0.201	0.156	0.233

Notes. Missing columns give the percentage without sSFR on each side. The observed difference uses classified galaxies; min/-max assign all missing galaxies on either side to quenched or star-forming, using all-galaxy denominators.

Extreme-assignment bounds for the missing class. Because the missing-sSFR class could in principle hide differential quenching, Table B.1 bounds every raw quenched-fraction contrast by the four extreme assignments in which *all* missing-sSFR galaxies on *each* side are set to quenched or to star-forming, with consistent all-galaxy denominators on both sides. The differential missingness is negligible against Control$_{4B}$ and Control$_{4C}$ (7% of CG$_4$ members versus 6% and 7%) and material only against RG$_4$ (2%). The raw RG$_4$ contrast survives all four corners (all-member difference between 0.12 and 0.21 around the observed 0.17; satellites 0.16 to 0.23), whereas the small Control$_{4B}$ and Control$_{4C}$ differences can change sign only under the doubly adversarial corner and are not the basis of any claim in this paper.

Star formation at fixed morphology. Table B.2 gives descriptive quenched fractions at fixed vote-based morphology. Table B.3 uses one common population—sSFR-classified galaxies with an E, S, or uncertain finite-vote morphology—for the total CG$_4$-minus-control difference and both Kita-

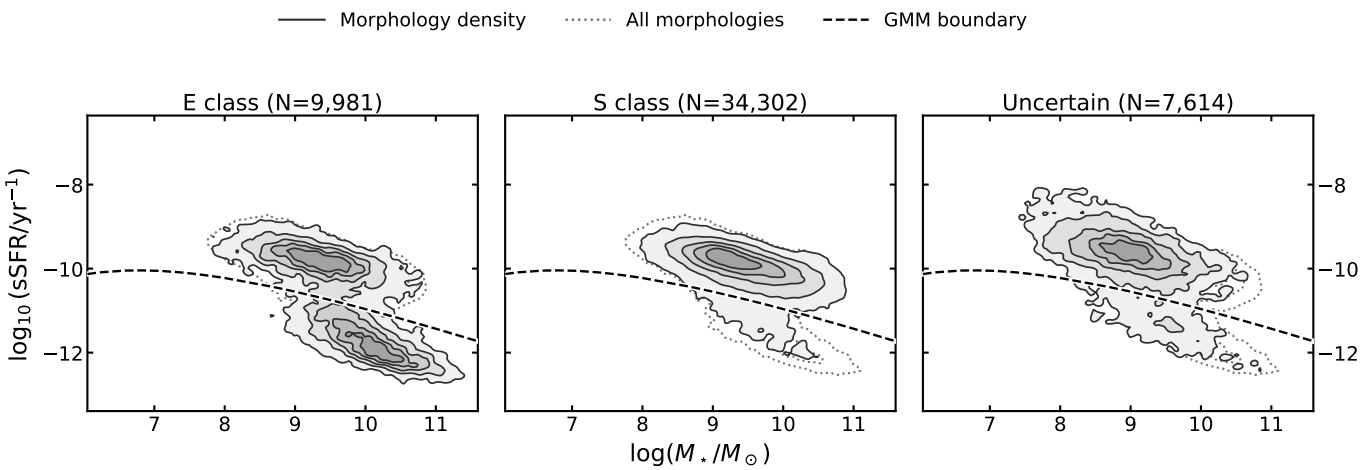


Fig. B.1. Galaxy Zoo morphology in sSFR–stellar-mass space for the SDSS non-AGN reference. Panels show the E, S, and uncertain classes. Solid class-specific contours enclose 20, 40, 60, 80, and 95% of the density after display-only Gaussian smoothing with $\sigma = 1.25$ histogram bins; grey dotted contours show the smoothed full reference distribution, and the dashed curve is the unchanged adopted GMM boundary.

Table B.2. Conditional quenched fractions

	CG_4	$Control_{4B}$	$Control_{4C}$	RG_4
GZ1 E	$0.908^{+0.040}_{-0.065}$	$0.903^{+0.016}_{-0.019}$	$0.887^{+0.017}_{-0.020}$	$0.850^{+0.069}_{-0.111}$
Spiral	$0.326^{+0.105}_{-0.090}$	$0.357^{+0.027}_{-0.026}$	$0.328^{+0.027}_{-0.026}$	$0.295^{+0.084}_{-0.072}$

Notes. For the uncertain-morphology class, $P(\mathrm{Q}|\mathrm{U}) = 0.429$, 0.552, 0.594, and 0.267 ($n = 7$, 116, 143, and 15 classified galaxies, same sample order).

gawa components. Its intervals use 4000 coupled resamples of physical Lim hosts; a host shared across comparison sides receives the same resampling weight.

Table B.3. Quenched-fraction decomposition on a common morphology-available population

Control	$N_{CG_4}/N_{\text{control}}$	Total Δf_Q (95% CI)	Morphology mix (95% CI)	Conditional term (95% CI)
Control_{4B}	213/2470	+0.049 [−0.023, +0.117]	+0.066 [+0.029, +0.102]	−0.017 [−0.073, +0.039]
Control_{4C}	213/2476	+0.064 [−0.010, +0.130]	+0.062 [+0.024, +0.100]	+0.002 [−0.056, +0.056]
RG_4	213/204	+0.201 [+0.105, +0.303]	+0.152 [+0.102, +0.208]	+0.050 [−0.035, +0.136]

Notes. Every displayed term uses the same population and total = morphology mix + conditional term algebraically. The corresponding differences among all sSFR-classified galaxies, which use different denominators because morphology need not be available, are +0.025, +0.040, and +0.174; they are reported only to reconcile the two populations. Intervals are percentile intervals from a physical-host bootstrap, not equivalence bounds.

Appendix C: Compact groups as projected cores of ordinary groups

The compact-group exclusion of Sect. 2.1 is not a technicality: for 61 of the 765 parent Lim groups, the would-be Control_{4C} quartet (BGG plus the three closest eligible projected companions) contains at least one CG_4 galaxy—233 CG_4 galaxies in total. Overlap is not the same as identity. Of the 56 non-isolated compact groups, 51 share at least one member with the would-be host core, but only 46 are exact four-member identities; 4 share three members and 1 shares one. Exact identity holds for 34 of 37 predominant and 12 of 19 embedded systems. Figure 1 shows one exact identity and one off-centre case. A control candidate is removed if its selected quartet contains a CG_4 galaxy; the rule does not remove every quartet from a host that contains some other CG_4 member, and 3 such Control_{4C} hosts are legitimately retained. The isolated entries in Table C.1 are trivial self-overlap: their Lim host is the compact group itself.

Appendix D: Statistical conventions

Independence unit. The unit for every inferential contrast among the group catalogues is the *physical* Lim group: standard errors are clustered by Lim group (unifying the overlapping control definitions, with CG_4 systems assigned to their host Lim group, or to their own identifier when they have no Lim counterpart). The decomposition resamples physical hosts jointly across comparison sides, and matched-quartet inference resamples connected components of the bipartite treated-quartet/control-host graph. Pooled models across the three controls are secondary summaries computed after deduplicating the overlapping control samples to one row per physical galaxy, with the kept label following the priority $\text{RG}_4 > \text{Control}_{4B} > \text{Control}_{4C}$ (Appendix G). Purely descriptive reference-sample diagnostics report their own unit where they appear.

Table C.1. Compact-group overlap with would-be Control_{4C} quartets

CG_4 class	CG_4 groups	Lim groups	Galaxies
Embedded	14 / 19	14	52
Isolated	6 / 6	6	24
Predominant	37 / 37	37	145
Split	4 / 16	4	12

Notes. "CG_4 groups" counts compact groups with at least one member inside a parent-group quartet, over the total number of compact groups of that class (split groups included, as in the exclusion).

Tests and p-values. Descriptive comparisons use two-sided Fisher exact tests for 2×2 tables and rank-sum or bootstrap comparisons for distributions and median differences; Spearman and Pearson coefficients are diagnostics of monotonic or linear trends only. Paired sign-flip permutation p-values use add-one tail counts, $p = (k + 1)/(B + 1)$ with k the number of permutations at least as extreme as the observation; bootstrap sign-crossing p-values are $2\min[(k_{\le 0} + 1)/(B + 1), (k_{\ge 0} + 1)/(B + 1)]$, capped at unity, where $k_{\le 0}$ and $k_{\ge 0}$ count resampled differences on either side of zero; displayed p-values are floored at 10^{-4}. Adjusted odds ratios come from binomial generalised linear models with cluster-robust covariance; continuous covariates are standardised, so coefficients of continuous terms are per standard deviation of the fitted frame. Multiplicity control is reserved for exploratory screens: Holm adjustment for the colour, measurement-availability, and magnitude-gap screens, and Benjamini–Hochberg false-discovery-rate control for the dominance, morphology–environment, and second-order screens.

Covariates. The velocity dispersion σ_v that enters models and matching is the quartet-level gapper estimate (Wainer & Thissen 1976) on the four selected members' redshifts, converted as $c\,\sigma_z/(1+\bar{z})$, for *all* samples, including Control_{4B} and Control_{4C} quartets embedded in richer hosts (this is how the spurious Lim 3688 quartets of Sect. 2.1 reach $\simeq 2000\,\text{km s}^{-1}$). With four tracers the estimator is intrinsically noisy: in a Gaussian Monte Carlo of 10^5 quartets, $\sigma_{\text{gapper}}/\sigma_{\text{true}}$ has median 0.96 with a 16–84% range of 0.57–1.42. A candidate covariate enters an adjusted model when it is non-missing for at least 65% of that model's frame (matching variables require 70%); in practice σ_v, group luminosity, stellar mass, redshift, and rank are complete for every quartet and always enter, and no harmonised group-mass covariate is available (Appendix G.3).

Overlap coefficient. Where two distributions are compared for common support (compactness in Sect. 2.1, $\log T_i$ in Sect. 3.4), the overlap coefficient is the integral of the minimum of the two normalised histograms on common bins, between 0 (disjoint) and 1 (identical).

Appendix E: Main-sequence and quenched-sequence details

We model the star-forming main sequence as a function of stellar mass with polynomials of order 1–4 fitted to the star-forming galaxies of the SDSS non-AGN reference sample; the diagnostic is the residual RMS scatter, not a for-

mal reduced χ^2, because no per-galaxy uncertainty model is used. We adopt order 2, the lowest order with the smallest five-fold cross-validated RMS (0.305 dex, against 0.306 dex for order 1), and compute the residual of each star-forming galaxy with respect to it; a positive residual means a higher sSFR at fixed stellar mass than the reference relation. In Sect. 3.2, Δ is the CG_4-minus-control median residual with the central 68% group-bootstrap interval on the median difference and the two-sided sign-crossing probability (10 000 resamples). The quenched sequence shown in the online supplement is obtained with the same procedure on the 7705 quenched galaxies of the reference sample: order 4 has the smallest cross-validated RMS (0.331 dex), and the residuals are attached to the quenched galaxies of the four samples.

Within CG_4, the star-forming main-sequence offsets of the embedded, isolated, and predominant classes are statistically consistent with one another.

Appendix F: Secondary and environmental diagnostics

This appendix collects the secondary diagnostics summarised in Sect. 3.5.

Magnitude-gap, BGG-distance, dominance, and colour diagnostics are provided in the online supplement; Paper I contains the fuller group-scale comparison.

F.1. Projected phase space

We ask whether the CG_4–RG_4 satellite contrasts vary with projected position relative to the BGG, using 174 CG_4 and 166 RG_4 satellites ranked by projected BGG-centric distance and, when available, $|\Delta v_{BGG}|/\sigma_v$; such coordinates are statistical proxies for infall state, not orbital reconstructions (Mahajan et al. 2011; Oman & Hudson 2016; Wetzel et al. 2013).

In the innermost satellite-distance bin, the quenched fractions are 68.3% for CG_4 and 48.2% for RG_4, a difference of 20.1 percentage points; the corresponding GZ1 E-class fractions are 64.7% and 35.6% (difference 29.1 points).

The quenched-fraction contrast is similar between the inner and outer distance ranks, so the data do not isolate the CG_4 offset to the immediate BGG environment.

After adjusting for stellar mass, redshift, distance bin, and available group-scale controls, the quenched-status model gives a CG_4 odds ratio of 1.18 (95% CI 0.43–3.24; $p = 0.742$), and the analogous E-class model gives 2.42 ($p = 0.078$).

A nearest-neighbour mass–redshift matched check on 126 CG_4–RG_4 satellite pairs gives matched quenched- and E-class-fraction differences of 9.5 and 18.7 percentage points.

In compact groups the member counts are small, velocity dispersions are noisy, and projection mixes recently accreted and long-resident satellites, so these bins locate where the contrast appears; they are not a dynamical clock.

Group radii and crossing times use projected physical separations from the adopted Planck 2015 angular-diameter distance. This corrects the luminosity-distance convention used for that diagnostic in Paper I, decreasing the values by 3.3–9.6% (median 7.2%) without changing the reported correlations at quoted precision. The optical-colour offsets remain control dependent; full diagnostics are in the online supplement.

F.2. Continuous group-scale descriptors

As an exploratory complement to the class comparison, Table F.1 fits satellite E-versus-S classification against four continuous group-scale descriptors: $\log\langle R_{ij}\rangle$, σ_v, $f_{L,BGG}$, and $\log t_{cross}$, adjusting for stellar mass and clustering by group. Descriptors are standardised within sample. Within CG_4, only compactness is associated with the satellite mix: smaller $\langle R_{ij}\rangle$ corresponds to more E-class satellites (OR 0.65, 95% CI [0.50, 0.84], $p_{BH} = 0.005$); the other descriptors are weak.

Table F.1. Continuous group-scale screen

Descriptor	CG_4	$Control_{4B}$	$Control_{4C}$	RG_4
$\log\langle R_{ij}\rangle$	0.65 [0.50, 0.84] p_{BH}=0.005	0.80 [0.72, 0.89] $p_{BH}=1.2\times10^{-4}$	0.70 [0.63, 0.78] $p_{BH}=<10^{-4}$	0.92 [0.64, 1.34] p_{BH}=0.757
σ_v	0.90 [0.65, 1.25] p_{BH}=0.736	1.03 [0.93, 1.15] p_{BH}=0.736	1.21 [1.09, 1.35] p_{BH}=0.002	0.90 [0.61, 1.32] p_{BH}=0.736
$f_{L,BGG}$	1.19 [0.88, 1.59] p_{BH}=0.591	1.02 [0.90, 1.16] p_{BH}=0.784	1.06 [0.95, 1.19] p_{BH}=0.591	1.19 [0.79, 1.81] p_{BH}=0.736
$\log t_{cross}$	0.88 [0.66, 1.17] p_{BH}=0.736	0.87 [0.78, 0.97] p_{BH}=0.030	0.73 [0.65, 0.81] $p_{BH}=<10^{-4}$	0.99 [0.69, 1.43] p_{BH}=0.978

Notes. Odds ratios and 95% intervals are per within-sample standard deviation, adjusted for stellar mass. p_{BH} denotes Benjamini–Hochberg adjustment over the exploratory screen. N = 157 CG_4 satellites in 59 groups; control counts are given in the released table.

Appendix G: Pooled models, matching details, and selection diagnostics

G.1. Pooled adjusted models

As a compact summary across the heterogeneous control definitions we also fit binomial models to the pooled catalogue. Because the control samples overlap heavily, the pooled control side is first deduplicated by SDSS objid and standard errors are clustered by physical Lim group; the common adjustment set includes stellar mass, redshift, satellite/BGG rank, group luminosity, and velocity dispersion. This pooled result does not replace the per-control contrasts. For quenched status, the CG_4 odds ratio is 1.35 (95% CI [0.98, 1.86]; p = 0.068). For quenched satellite status, the CG_4 odds ratio is 1.53 (95% CI [1.10, 2.11]; p = 0.011). For GZ1 E-class morphology, the CG_4 odds ratio is 1.77 (95% CI [1.32, 2.37]; $p = 1.4\times10^{-4}$). For GZ1 E-class satellite morphology, the CG_4 odds ratio is 2.08 (95% CI [1.51, 2.88]; $p < 10^{-4}$). For GZ1 E-class BGG morphology, the CG_4 odds ratio is 0.78 (95% CI [0.39, 1.53]; p = 0.464). The pooled coefficients reinforce the satellite morphology shift; the corresponding forest plot is provided in the online supplement. The quenched estimate changes across the separate controls and is not reproduced by the matched contrast.

G.2. Galaxy-level matching details

The galaxy-level control pool of Sect. 3.1 is deduplicated by SDSS objid before matching: 5828 control rows collapse to 3609 unique galaxies. The propensity model uses $\log M_\star$, redshift, rank, $\log L_{group}$, and σ_v, and the match is one-to-one without replacement within exact rank strata. The 14 unmatched CG_4 galaxies lay outside the common propensity-score support or exceeded the caliper. The maximum absolute covariate standardized mean difference falls from 0.37 to 0.08 (Fig. G.1). By kept label, the matched controls comprise 159 $Control_{4B}$, 57 $Control_{4C}$, 18 RG_4; physically, 18 of the 234 matched controls are RG_4 galaxies (which also appear in $Control_{4B}$ through the sample overlap), and the per-galaxy origin of every matched control is released with the derived tables. Confidence intervals and p-values come from a bootstrap blocked by the treated galaxy's compact group.

The 234 matched controls occupy 204 physical Lim groups, and 26 control hosts link more than one treated

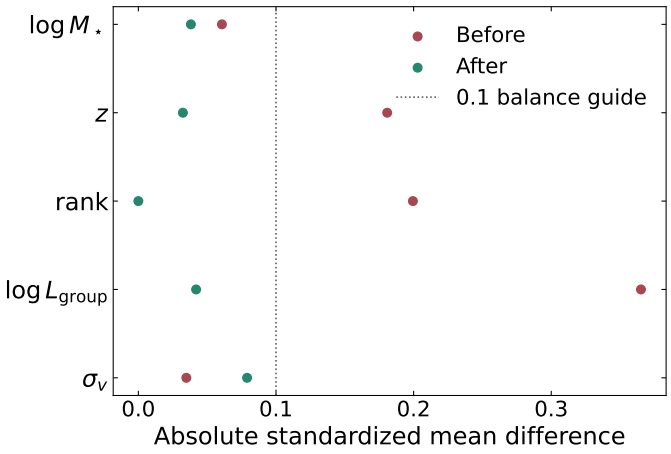


Fig. G.1. Matching balance. Absolute standardized mean differences before and after matching; the labelled 0.1 line is a conventional diagnostic guide.

CG_4 group. Resampling the 33 connected components of the treated-group/control-host graph instead, the E-class fraction difference remains 0.139 (95% interval [0.051, 0.244]; p = 0.004), and the quenched-fraction difference is 0.031 ([−0.090, 0.120]; p = 0.649). The analogous matched-size sensitivities are −0.019 dex for the Simard radius (p = 0.310), −0.033 dex for the Petrosian radius (p = 0.016), and −0.008 for the concentration (p = 0.896).

G.3. Selection diagnostics

The fractions with complete measurements for the major derived quantities are shown in Fig. G.2, including the Simard and Petrosian size availability used in Sect. 3.5 and the D_n4000/Hδ_A availability of the spectral indices. The median nearest-neighbour angular separation is 100.9 arcsec in CG_4 and 255.9 arcsec in the controls; the fractions below the 55-arcsec SDSS fibre-collision scale are 19.4% and 7.7%. No major availability difference has both an adjusted p-value below 0.05 and a difference exceeding 10 percentage points.

The group-scale covariates are radius scale, velocity dispersion, group luminosity, and BGG luminosity dominance. The tables do carry an M_{200c} estimate, but it is the Yang–Lim abundance-matching M_{180m} of the identified *host* halo

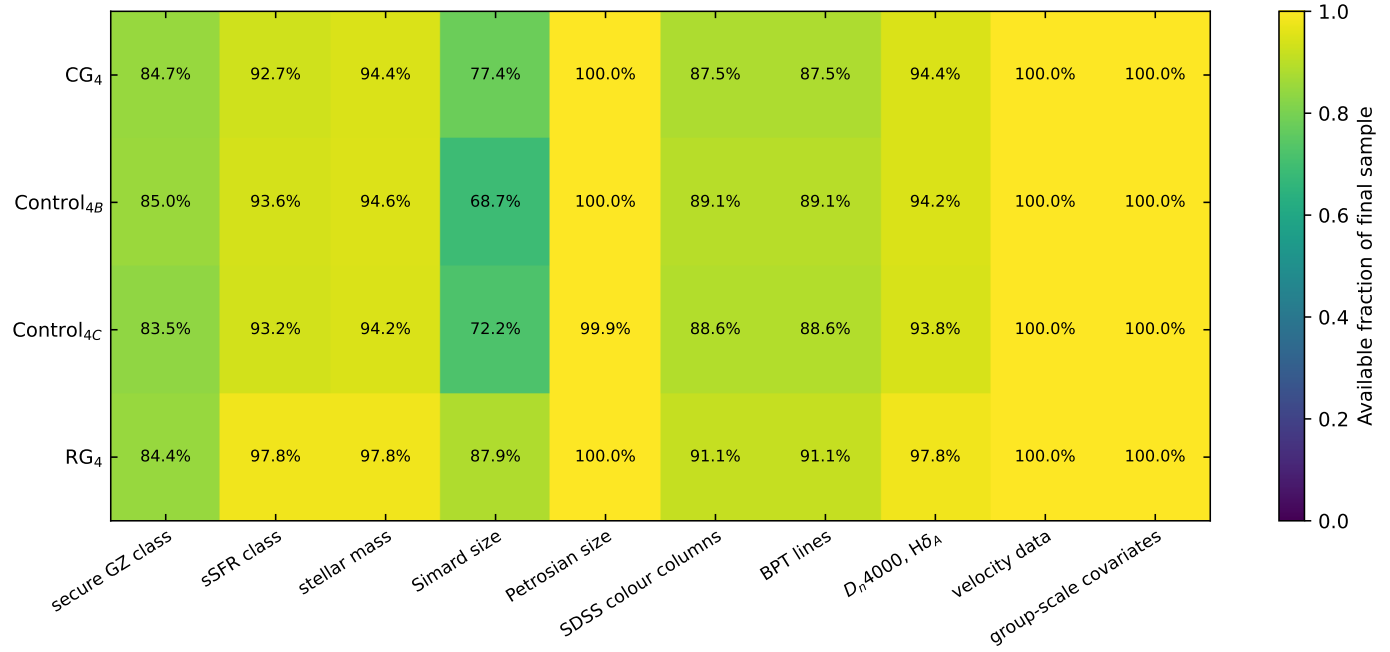


Fig. G.2. Measurement availability. Cells give percentages of the final per-sample galaxy count. The colour row describes the four derived SDSS columns, the spectral-index row requires finite non-sentinel values with positive errors, and the Simard row includes all size-quality cuts.

(converted as in Paper I, Appendix A), not a quartet property: for embedded quartets it describes the richer host while for isolated systems it describes the quartet itself, it derives from group luminosity by abundance matching and is therefore nearly collinear with the group-luminosity covariate, and it is entangled with the compact/loose selection. We therefore exclude it from the harmonised covariate set. A sensitivity fit adding $\log M_{200c}$ to the published covariates leaves every E-class conclusion unchanged: the pooled odds ratio moves from 1.77 to 1.62 ($p = 0.0020$), and the per-control odds ratios become 2.00, 1.56, and 2.65 against $\mathrm{Control}_{4B}$, $\mathrm{Control}_{4C}$, and RG_4 (all $p < 0.005$).

In the catalogue-row sensitivity fit, restricting the pairwise tidal index of Sect. 3.4 to quartet co-members is innocuous: recomputing T_i with all spectroscopic Lim members of the identified host group included (positions from the group catalogue; stellar masses from MPA–JHU where available, so the host-inclusive index is itself a lower bound) shifts $\log T_i$ by a median of 0.009 dex (sample medians ≤ 0.02 dex, and identically zero for RG_4, whose groups have exactly four members), with rank correlation 0.91 between the two definitions, and refitting the with-index E-class model on the host-inclusive index moves the attenuated CG_4 coefficient only from 1.39 to 1.33. T_i is thus the local, galaxy-scale tidal density traced by catalogued galaxies; any smooth host-halo contribution is probed separately through the host-mass sensitivity fit above and the within-host analysis of Sect. 3.1. The colour-selection mass diagnostics are provided in the online supplement.

Appendix H: Robustness details

Sample-definition sensitivities. Restoring the 16 Zheng–Shen split systems while retaining the published model definitions leaves the morphology result qualitatively unchanged: the adjusted all-galaxy E-class odds ratios against $\mathrm{Control}_{4B}$, $\mathrm{Control}_{4C}$, and RG_4 are 1.97 (95% CI [1.46, 2.64]), 1.49 ([1.12, 1.96]), and 2.62 ([1.69, 4.06]), and the corresponding satellite odds ratios are 2.57 ([1.88, 3.50]), 1.84 ([1.38, 2.46]), and 2.96 ([1.75, 5.01]). Retaining the anomalous Lim 3688 quartets likewise leaves the coefficients essentially unchanged: the all-galaxy and satellite odds ratios are 2.18 and 2.65 against $\mathrm{Control}_{4B}$, and 1.68 and 1.94 against $\mathrm{Control}_{4C}$.

Crowding. Galaxy Zoo morphology is image-based, so crowded closest pairs could bias the votes. After excluding galaxies whose nearest projected neighbour within the catalogue group lies within 55 arcsec (retaining 200 CG_4 and 2534 $\mathrm{Control}_{4C}$ galaxies), the CG_4 all-row E-class fraction *rises* relative to the full-sample value of Table 3 (52.5% against 40.1% for $\mathrm{Control}_{4C}$, all retained rows including uncertain and missing classes), and the CG_4–$\mathrm{Control}_{4C}$ contrast remains clear. In the joint model, the close-neighbour flag has OR 1.98 ($p = 6.1 \times 10^{-4}$), while the CG_4 coefficient at fixed flag status is OR 1.76 ($p = 2.6 \times 10^{-4}$); the former compares close with non-close galaxies, the latter CG_4 with controls at fixed flag status, so neither contradicts the rising raw CG_4 fraction after exclusion. The checks do not prove immunity to crowding, but they rule out the closest-pair artefact as the sole origin of the result.

Without σ_v. Refitting the per-control and pooled models without the σ_v covariate, on identical samples and with identical clustering, moves the E-class odds ratios from 2.14 to 2.22 ($\mathrm{Control}_{4B}$), 1.67 to 1.68 ($\mathrm{Control}_{4C}$), and 2.81 to 2.76 (RG_4), less than their standard errors. The galaxy-level match retains the same 234 CG_4 galaxies with an E-class fraction difference of 0.16. The per-control group-level differences become $\Delta = 0.28$, 0.10, and 0.25 against $\mathrm{Control}_{4B}$, $\mathrm{Control}_{4C}$, and RG_4 (permutation $p = 0.0001$,

0.056, 0.0016). The $\mathrm{Control}_{4B}$ change reflects the composition of the matched set rather than the CG_4 side: the with- and without-σ_v matches share only 7 of 54 control groups, the matched-control mean E-class satellite fraction changes by −0.12 while the CG_4 mean is unchanged, and without σ_v the matched controls are dynamically cooler than their CG_4 partners (residual σ_v standardized mean difference 0.27, medians 160 against 129 $\mathrm{km\,s^{-1}}$) and less well balanced in redshift, so the with-σ_v match remains the primary one and the two values bracket the $\mathrm{Control}_{4B}$ contrast.

Within-host estimators. The conditional-logit sample of Sect. 3.1 contains 791 galaxies in 53 host strata. A binomial GLM with host fixed-effect dummies and cluster-robust standard errors gives OR 1.57 (95% CI [1.02, 2.43]; $p = 0.041$); we prefer the conditional logit because it eliminates the host intercepts rather than estimating many indicators from small strata. The within-host quenched odds ratio is 0.75 (95% CI [0.45, 1.25]; $p = 0.269$; 915 complete cases), and 1.22 without the radius covariate ($p = 0.412$). A host-BGG alignment check supports the satellite interpretation: 56 of 62 non-split CG_4 systems have the same quartet and Lim-host BGG, the 6 misaligned systems are embedded systems whose host BGG is brighter than the compact-group BGG, and all 186 quartet-rank CG_4 satellites are Lim-host satellites; restricting CG_4 to the 56 BGG-aligned systems leaves the three satellite E-class contrasts stable (against $\mathrm{Control}_{4C}$, OR 1.98).

DS18 classification. The DS18 cross-match of Sect. 2.2 covers 173/248 (70%) of the CG_4 galaxies and 2263/3609 (63%) of the deduplicated controls, i.e. the DR7-era population; it is not worse in CG_4, but neither rate establishes representativeness. With the DS18 $T\mathrm{Type} \leq 0$ outcome and the per-control covariates, standardisation, and physical-group clustering of Sect. 3.1, the CG_4 coefficients are positive in all three contrasts (Table 6); support is broad except that 83% of CG_4 values lie within the RG_4 group-luminosity range, limiting that smallest comparison. Because DS18 P_{S0} separates S0 from E only within $T\mathrm{Type} \leq 0$, the environment dependence of the E/S0 mix is left to a dedicated analysis.

Appendix I: Extended size diagnostics

The size-availability counts are included in Fig. G.2; the descriptive mass–size relations and the per-sample sizes predicted at $\log(M_\star/M_\odot) = 10.3$ are provided in the online supplement. The concentration index $C = R_{90,r}/R_{50,r}$ provides a model-independent structural proxy: the adjusted satellite coefficient is 0.068 ($p = 0.124$), and the matched concentration difference is −0.008 ($p = 0.904$), offering no evidence for a fixed-mass concentration shift.

The crowding checks mirror Appendix H: excluding galaxies with a projected neighbour within 55 arcsec leaves a satellite coefficient of −0.013 dex ($p = 0.463$), and adding the close-neighbour flag as a covariate gives −0.011 dex ($p = 0.479$).

The per-galaxy difference between the two measures, $d = \log R_{e,r} - \log R_{50,r}$, averages 0.054 dex, and is larger for GZ1 E-class galaxies (0.074 dex) than for S-class galaxies (0.042 dex), as expected if the Petrosian aperture preferentially truncates concentrated profiles. Its rank correlation with the nearest-neighbour separation is $\rho_S = 0.012$ ($p = 0.524$), with no detected monotonic relation to neighbour separation.

Adding the pairwise projected tidal index of Sect. 3.4 to the all-galaxy size model changes the CG_4 coefficient from −0.012 dex to 0.005 dex on the pairwise complete-case sample; the local projected-density proxy attenuates the size coefficient in the same way it attenuates the morphology coefficient.

The radial diagnostic is exploratory: mass- and redshift-adjusted satellite size residuals correlate with projected BGG-centric distance with $\rho_S = 0.17$ ($p = 0.026$) in CG_4 and $\rho_S = 0.18$ ($p < 10^{-4}$) in the pooled controls. A more negative inner-region residual would be the tidal-truncation expectation; the observed trends are similar in CG_4 and the controls, so the data do not isolate a CG_4-specific inner-region size suppression.

Within morphology classes, the satellite-only fits give 0.036 dex for GZ1 E-class satellites ($p = 0.068$) and −0.005 dex for S-class satellites ($p = 0.831$), and the CG_4×morphology interaction is 0.032 dex ($p = 0.348$). These strata are secondary diagnostics of whether size carries information beyond the morphology shift.